B. TECH-ME-III-I Sem

23A03502P	THERMAL ENGINEERING LAB	L	T	P	C
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Course objectives: The objectives of the course are to	
1	To impart knowledge on working principles of various thermal equipments like compressors, IC Engines, Boilers etc.,
2	To study the working principle of IC engines, performance and characteristics in terms of heat balancing, economical speed variations, air fuel ratio etc.,

List of Experiments:

1. Valve / Port Timing Diagrams of an I.C. Engines
2. Performance Test on a 4 -Stroke Diesel Engines
3. Performance Test on 2-Stroke Petrol engine
4. Evaluation of Engine friction by conducting Morse test on 4-Stroke Multi cylinder Engine
5. Retardation and motoring test on 4- stroke engine
6. Heat Balance of an I.C. Engine.
7. Air/Fuel Ratio and Volumetric Efficiency of an I.C. Engines.
8. Performance Test on Variable Compression Ratio Engines, economical speed test.
9. Performance Test on Reciprocating Air – Compressor Unit
10. Study of Boilers
11. Dismantling / Assembly of Engines to identify the parts and their position in an engine.
12. Exhaust Emission test on IC Engines.

Course Outcomes: On successful completion of the course, the student will be able to,	
CO1	Performance Test on 4-Stroke Diesel and 2-Stroke Petrol engine.
CO2	Able to evaluate the Engine friction of 4-Stroke Multi cylinder Engine and Air/Fuel ratio and Volumetric efficiency of I.C.Engines.
CO3	To calculate the heat balance of the IC Engines.
CO4	To calculate the efficiencies and performance characteristics of the engines.
CO5	Study the boilers and identify parts of the engine parts.

Online Learning Resources:

- https://www.youtube.com/watch?v=i4SF47hjnjQ&list=PL0AQx5JlTK3WUCXXkA9Hev3FFLz4sESSg
- https://www.youtube.com/watch?v=B-rFIdOi-No&list=PLkUEX3IbW7lfdC2ieft_9FH5zAAvUfZAn